

ENGINEERING REBOOT 2026

Week-02 Master Workbook v5.0 (Locked Edition)

Part 2 - Day-06 Engineering Lab Manual

Mission Brief

Software becomes useful when it can make decisions. Day-06 introduces decision making using if, elif, and else statements.

Engineering Context

Decision making is used in building automation, banking systems, robotics, industrial controls, embedded systems, and web applications.

Learning Objectives

Understand if, elif, else; evaluate conditions; execute alternate paths; design simple decision trees.

Daily Folder Structure

Week-02/Day-06 contains Exercises, Projects, Screenshots, Notes/Notes.md, and Journal/Journal.md.

Required Deliverable

decision_program.py containing the Age Classifier and Temperature Advisor exercises.

VS Code File Creation Procedure

Open VS Code, navigate to Exercises, right-click, select New File, and create decision_program.py.

Exercise 01 - Age Classifier

Classify age ranges: Child (0-12), Teenager (13-17), Adult (18-64), Senior (65+). Validation tests T01-T04.

Exercise 02 - Temperature Advisor

Provide recommendations based on temperature. Validation tests T01-T03.

Expected End-State

All required screenshots, Notes.md, Journal.md, and decision_program.py present in the Day-06 folder structure.

Notes Activity

Complete Notes.md using the locked template and document concepts learned.

Journal Activity

Complete Journal.md using the locked template and answer reflection questions.

Git Workflow

git status, git add ., git commit -m 'Day 06 - Completed exercises and documentation', git push.

Acceptance Criteria

All exercises complete, evidence collected, notes and journal completed, commit and push completed.

Stretch Goal

Reject negative ages and display 'Invalid Age Entered'.